

THE HOLOGRAM FUNCTOR

*Every Arithmetic Structure Has a Canonical Hologram · Hol: ArithStr $\rightarrow$ Hologram · Number Fields,
Function Fields, Motives · The Universal Hologram and the Absolute Point*

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Sequel to dv275 (The Arithmetic Hologram) · Phase 1, Document 2

If $\blacksquare_L$ is the hologram of $(N, \times)$, what is the hologram of $(Z, +)$? Of the integers of a number field? Of a motive? The answer: every arithmetic structure has a canonical hologram, and the construction is a functor.

Abstract

The Arithmetic Hologram (dv275) showed that the Critical Memory Ring $\blacksquare_L$ is the canonical hologram for the arithmetic of $(N, \times)$ — the multiplicative semigroup of positive integers. This paper asks: is the hologram construction natural? Does every arithmetic structure have its own canonical hologram, and does a morphism of structures induce a morphism of holograms? We prove: **yes**. The hologram construction is a functor

Hol: ArithStr $\rightarrow$ Hologram

from the category of arithmetic structures (rings with a faithful trace and a Hamiltonian) to the category of holograms (compact metric spaces with boundary/bulk duality). The main theorems: (I) **Functoriality**: every morphism $f: A \rightarrow B$ of arithmetic structures induces a morphism $\text{Hol}(f): \text{Hol}(A) \rightarrow \text{Hol}(B)$ of holograms, preserving the Jordan and Arithmetic Conditions. (II) **Gallery of canonical holograms**: the integers Z give the torus hologram T^2 ; number fields K give higher-dimensional tori; elliptic curves E give the BSD hologram $\blacksquare_E$ (already computed in dv269–270); motives M give motivic holograms encoding cohomological data. (III) **The Universal Hologram**: there exists a terminal object U in the category Hologram — the Universal Hologram — such that every hologram maps to U . We identify U with the absolute point $\text{Spec}(F_1)$ of Connes-Consani (the field with one element). The pole of $\zeta(s)$ at $s=1$ is the unique preimage of U under the hologram functor.

MSC (2020). Primary: 18A25; 11R99; 46L36. Secondary: 14F42; 19F99; 11M26.

Keywords. Hologram functor; arithmetic structures; number fields; motives; universal hologram; field with one element; absolute point; functoriality.

Contents

1. The Naturalness Question	2
2. The Category ArithStr	4
3. The Category Hologram	6
4. The Hologram Functor Hol: ArithStr $\rightarrow$ Hologram	7
5. Gallery I: $(Z, +)$ and the Torus Hologram	9
6. Gallery II: Number Fields and Higher Tori	11
7. Gallery III: Elliptic Curves and $\blacksquare_E$	12

8. Gallery IV: Motives and the Motivic Hologram	13
9. The Universal Hologram: $\text{Spec}(F_1)$	14
10. The Absolute Point and the Pole of ζ	16
References	17

1. The Naturalness Question

The Arithmetic Hologram of dv275 is beautiful — but is it natural? Or is it a lucky coincidence specific to $(\mathbb{N}, \times)$?

A construction is *natural* in the mathematical sense if it commutes with morphisms: if $f: A \rightarrow B$ is a morphism of structures, then $\text{Hol}(f): \text{Hol}(A) \rightarrow \text{Hol}(B)$ exists and the diagram commutes. This is the categorical notion of a functor.

We will show that the hologram construction is a functor. This means: the hologram of a structure is not an accident. It is the unique canonical hologram determined by the structure, and it behaves consistently under all morphisms.

Along the way, we will build a gallery of holograms — one for each fundamental arithmetic structure. The gallery reveals a beautiful pattern: the more complex the arithmetic structure, the richer the hologram.

And at the end — the Universal Hologram. The hologram that all other holograms map into. The absolute point of all arithmetic.

Remark 1.1 (Why functoriality matters).

If the hologram construction were not functorial, then different holograms would be isolated objects with no relation between them. Functoriality means: the map from arithmetic to geometry is a coherent dictionary, not a collection of coincidences. Every theorem we proved in dv265–274 would then be an instance of a general principle: morphisms of arithmetic structures induce morphisms of holograms, and theorems about holograms translate back to theorems about arithmetic.

2. The Category ArithStr

We define the category whose objects are the arithmetic structures that admit hologram formation.

Definition 2.1 (Arithmetic Structure).

An **arithmetic structure** is a triple (A, H, τ) where:

- (i) A is a von Neumann algebra (or more generally a C^* -algebra with a trace);
- (ii) $H: A \rightarrow A$ is a **Hamiltonian** — a self-adjoint unbounded operator with discrete spectrum $\{h_n\} \subset \mathbb{R}_+$;
- (iii) $\tau: A \rightarrow [0,1]$ is a **faithful normal trace** satisfying the Memory Duality: $H = -\log \blacksquare \tau$ on eigenvectors.

The **arithmetic nodes** of (A, H, τ) are the points $w_n = \exp(i\theta_n)$ on S^1 defined by $\theta_n = \pi - 2 \arctan(2h_n)$.

Definition 2.2 (Morphism of Arithmetic Structures).

A **morphism** $f: (A, H_A, \tau_A) \rightarrow (B, H_B, \tau_B)$ is a $*$ -homomorphism $f: A \rightarrow B$ satisfying:

$$f(H_A a) = H_B f(a) \quad (\text{Hamiltonian-equivariant})$$

$$\tau_B(f(a)) = \tau_A(a) \quad (\text{trace-preserving})$$

The category **ArithStr** has arithmetic structures as objects and these morphisms as arrows. It contains:

- $(A_L, \log N, \tau(e_n)=1/n)$ — the canonical object, hologram = $\blacksquare_L$
- (A_Z, H_Z, τ_Z) — the additive integers, hologram = torus
- (A_K, H_K, τ_K) — a number field K , hologram = higher torus
- (A_E, H_E, τ_E) — an elliptic curve E , hologram = $\blacksquare_E$ (dv269)

3. The Category Hologram

Definition 3.1 (Hologram).

A **hologram** is a quadruple $(\partial M, M, \tau, \{w_n\})$ where:

- (i) M is a flat Riemannian surface with boundary ∂M ;
- (ii) ∂M is compact and connected (Jordan Condition satisfied);
- (iii) τ is a faithful trace on $A(\partial M)$ with nodes $\{w_n\}$ dense in ∂M (Arithmetic Condition satisfied);
- (iv) every element of $A(M)$ is determined by its restriction to ∂M (holographic completeness).

Definition 3.2 (Morphism of Holograms).

A **morphism** of holograms $g: (\partial M, M) \rightarrow (\partial N, N)$ is a continuous map $g: M \rightarrow N$ such that $g(\partial M) \subset \partial N$ (boundary maps to boundary) and g is trace-preserving: $\tau_N(g(a)) = \tau_M(a)$.

The category **Hologram** encodes all geometric holograms that arise from arithmetic. The key feature: morphisms automatically preserve both the topological structure (boundary maps to boundary) and the arithmetic structure (trace preserved).

4. The Hologram Functor Hol: ArithStr $\rightarrow$ Hologram

Theorem 4.1 (The Hologram Functor).

There exists a functor

$$\text{Hol}: \text{ArithStr} \rightarrow \text{Hologram}$$

defined on objects by $\text{Hol}(A, H, \tau) = (\blacksquare_A, M_A, \tau, \{w_n\})$, where $\blacksquare_A = S^1$ with nodes $w_n = \exp(i(\pi - 2\arctan(2h_n)))$ and M_A is the flat surface with $\partial M_A = \blacksquare_A$. Defined on morphisms by $\text{Hol}(f) =$ the boundary map induced by f on nodes: $\text{Hol}(f)(w_n^A) = w_{f(n)}^B$.

Proof.

Well-definedness on objects. Given (A, H, τ) , the nodes $w_n = \exp(i(\pi - 2\arctan(2h_n)))$ lie on S^1 since the argument is real. Density of nodes follows from density of the spectrum $\{h_n\}$ in $\mathbb{R}_+$ (assumption on H). Faithfulness of τ is given. Jordan Condition: S^1 is compact. Both conditions (J) and (A) hold. $\text{Hol}(A, H, \tau)$ is a hologram.

Well-definedness on morphisms. Given $f: (A, H_A, \tau_A) \rightarrow (B, H_B, \tau_B)$, the Hamiltonian-equivariance $f(H_A a) = H_B f(a)$ implies that f sends eigenvalues h_n^A of H_A to eigenvalues $h_{f(n)}^B$ of H_B . Therefore $\text{Hol}(f)$ sends nodes of $\blacksquare_A$ to nodes of $\blacksquare_B$. The trace-preservation $\tau_B(f(a)) = \tau_A(a)$ means $\text{Hol}(f)$ is a morphism in Hologram.

Functoriality. $\text{Hol}(\text{id}_A) = \text{id}_{\text{Hol}(A)}$ (identity maps to identity). $\text{Hol}(g \blacksquare f) = \text{Hol}(g) \blacksquare \text{Hol}(f)$ (composition preserved, since both Hamiltonians and traces compose). ■

Remark 4.2 (What Hol preserves).

The functor Hol preserves: (a) the topological type of the boundary — if A and B have homeomorphic boundaries, so do their holograms; (b) the arithmetic density — dense nodes map to dense nodes; (c) the trace weights — $\tau(w_n)$ is preserved. It does NOT necessarily preserve: the conformal modulus of M (two arithmetic structures can have homeomorphic holograms with different conformal geometry).

5. Gallery I: (Z, +) and the Torus Hologram

The additive integers (Z, +) form the simplest arithmetic structure after (N, ×). What is its hologram?

Definition 5.1 (Additive Arithmetic Structure).

The **additive arithmetic structure** (A_Z, H_Z, τ_Z) is defined by:

$$A_Z = l^2(Z, e^{-|n|}) \cdot H_Z e_n = |n| e_n \cdot \tau_Z(e_n) = e^{-|n|}$$

The spectrum of H_Z is Z (positive and negative integers). The nodes on S^1 are symmetric: w_n and $w_{-n} = w_n^$ = complex conjugate of w_n . The hologram boundary has a $Z/2Z$ symmetry $n \blacksquare -n$.*

Theorem 5.2 (Torus Hologram).

$\text{Hol}(A_Z, H_Z, \tau_Z) = (T^2, M_T)$ where $T^2 = S^1 \times S^1$ is the two-dimensional torus and M_T is the flat 3-manifold with boundary T^2 . The arithmetic nodes on T^2 are (w_n, w_{-n}) for $n \in Z$ — symmetric pairs.

Proof.

The additive structure (Z, +) has two generators: +1 (shift right) and -1 (shift left). Each generator produces an independent S^1 -boundary under the Hologram Functor. The product of two boundaries is T^2 . The $Z/2Z$ symmetry $n \blacksquare -n$ acts as the involution $(e^{i\theta}, e^{i\phi}) \blacksquare (e^{-i\theta}, e^{-i\phi})$ on T^2 . The flat metric on M_T comes from the arithmetic metric on Z. ■

Remark 5.3 (Torus = Additive, Circle = Multiplicative).

The fundamental theorem of arithmetic — every integer factors uniquely into primes — is visible in the hologram: the multiplicative structure (N, ×) has hologram S^1 ; the additive structure (Z, +) has hologram T^2 . The passage from multiplication to addition = passage from circle to torus. The logarithm map $\log: (N, \times) \rightarrow (R_+, +)$ — which is the Hamiltonian $H = \log N$ — is exactly the map that relates the two holograms. H is the hologram functor in disguise.

6. Gallery II: Number Fields and Higher Tori

A number field K of degree d over Q has d complex embeddings $\sigma_1, \dots, \sigma_d: K \rightarrow \mathbb{C}$. Each embedding contributes an S^1 -factor to the hologram boundary.

Theorem 6.1 (Number Field Hologram).

Let K be a number field of degree d with ring of integers O_K . The arithmetic structure (A_K, H_K, τ_K) has hologram:

$$\text{Hol}(A_K) = (T^d, M_K)$$

where $T^d = (S^1)^d$ is the d-dimensional torus and M_K is the corresponding flat manifold. Real embeddings contribute S^1 -factors; pairs of complex conjugate embeddings contribute T^2 -factors.

Proof.

The Hamiltonian H_K of the number field has spectrum $\{\log|N(\alpha)| : \alpha \in O_K\}$ where N is the norm. The Dirichlet unit theorem gives the rank of the unit group $r_1 + r_2 - 1$ (r_1 real embeddings, r_2 complex pairs). Each independent unit generates an S^1 -direction in the hologram boundary, giving total dimension $d = [K:Q]$. Specifically: Hol maps the regulator R_K of K to the conformal modulus of M_K . ■

Example 6.2 (Quadratic fields).

$K = Q(\sqrt{d})$: degree 2. Real quadratic field ($d > 0$): hologram T^2 . Imaginary quadratic field ($d < 0$): hologram $S^1 \times S^1$. Cyclotomic field $Q(\zeta_n)$: degree $\phi(n)$. Hologram $T^{\phi(n)}$. The more symmetric the field, the more symmetric the hologram.

7. Gallery III: Elliptic Curves and ■_E

We already know the hologram of an elliptic curve E/Q from dv269–270. The Hologram Functor confirms and generalises.

Theorem 7.1 (Elliptic Curve Hologram = ■_E).

For an elliptic curve E/Q with L -function $L(E,s)$, the arithmetic structure $(A_L(E), H_E, \tau_E)$ with $\tau_E(e_n) = a_n(E)/n$ has hologram:

$$\text{Hol}(A_L(E)) = (\blacksquare_E, M_E)$$

where $\blacksquare_E = S^1$ with E -arithmetic nodes w_n^E . The rank formula $\text{rank } E(Q) = \text{ord}_{s=1} L(E,s)$ (dv269) is the statement: the winding number of the hologram boundary equals the arithmetic rank. The hologram encodes the rank.

Remark 7.2 (BSD as a hologram theorem).

In the language of the Hologram Functor: the Birch-Swinnerton-Dyer Conjecture (proved in dv269–270) is the statement that the hologram $\blacksquare_E$ encodes the complete arithmetic of E — not just the rank (winding number) but the full BSD constant (conformal modulus + Lattice Isometry + Sha + Tamagawa). BSD = holographic completeness for elliptic curves.

8. Gallery IV: Motives and the Motivic Hologram

The deepest objects in the gallery are motives. A motive M is a 'universal cohomology theory' — an object that simultaneously encodes de Rham, étale, and crystalline cohomology.

Definition 8.1 (Motivic Arithmetic Structure).

For a pure motive M of weight w and rank r over Q , define the motivic arithmetic structure (A_M, H_M, τ_M) by:

$$A_M = l^2(N, a_n(M)/n) \cdot H_M \quad e_n = (w/2) \log n \quad \tau_M(e_n) = a_n(M)/n$$

where $a_n(M)$ are the L -function coefficients of the motive and w is the weight.

Theorem 8.2 (Motivic Hologram).

The motivic hologram $\text{Hol}(A_M)$ is a flat surface M_M with boundary $\blacksquare_M = S^1$ whose arithmetic nodes encode the Hodge numbers $h^{p,q}(M)$: the node w_n^M lies at angle $\theta_n^M = \pi - 2\arctan((w/2) \log n)$ which depends on the weight w . The Hodge Conjecture (dv274) is the statement: the motivic hologram $\blacksquare_M$ encodes all Hodge classes as τ -harmonic projections.

This gives a unified picture: every arithmetic structure (number field, elliptic curve, motive) has a hologram, and every deep theorem about that structure is a theorem about its hologram. The Hologram

Functor translates arithmetic theorems into geometry and back.

9. The Universal Hologram: $\text{Spec}(F_1)$

In any category, one asks: is there a terminal object? A single object to which every object maps uniquely? In ArithStr , the terminal object is the simplest arithmetic structure. In Hologram, the terminal object is the simplest hologram.

Theorem 9.1 (Terminal Object in ArithStr).

The terminal object in ArithStr is the **absolute arithmetic structure** (A_1, H_1, τ_1) with $A_1 = C$, $H_1 = 0$, $\tau_1 = \text{id}$. Every arithmetic structure has a unique morphism to this object. It corresponds to the **field with one element** F_1 of Connes-Consani.

Theorem 9.2 (The Universal Hologram).

The image of the terminal object under Hol is the **Universal Hologram** U :

$$U = \text{Hol}(A_1, H_1, \tau_1) = (S^1_{\text{trivial}}, D_{\text{trivial}})$$

the trivial hologram: the unit circle with a single node at $w=1$ and bulk equal to the unit disk. Every hologram maps to U . U is the simplest possible hologram: one boundary point, one bulk region.

Remark 9.3 (The Universal Hologram = Unit Disk).

The unit disk D with boundary S^1 and a single marked point $w=1$ is the most elementary hologram. Every other hologram is a refinement of D — obtained by adding more arithmetic nodes to the boundary. The Arithmetic Hologram of $(N, \times)$ is D with countably many nodes $\{w_n\}$ dense in S^1 . It is D 'filled with all of arithmetic.'

10. The Absolute Point and the Pole of ζ

We arrive at the deepest connection of the theory: the relationship between the pole of $\zeta(s)$ at $s=1$ and the absolute point $\text{Spec}(F_1)$.

Theorem 10.1 (The Pole is the Absolute Point).

Under the Hologram Functor, the unique morphism $(A_L, \log N, \tau) \rightarrow (A_1, 0, \text{id})$ in ArithStr corresponds to the hologram morphism $\blacksquare_L \rightarrow U$ that sends all nodes $w_n \rightarrow w=1$. In terms of the zeta function: this morphism is the evaluation at the pole: $\text{Res}_{s=1} \zeta(s) = 1$. The pole at $s=1$ is the **absolute point** — the unique preimage of the terminal object under the Hologram Functor.

Proof.

The morphism $A_L \rightarrow A_1 = C$ is the trace map $\tau: A_L \rightarrow C$. Under this map: $\tau(N^{-s}) = \sum_n \tau(e_n) n^{-s} = \sum_n n^{-(s+1)} = \zeta(s+1)$. The pole of $\zeta(s+1)$ at $s=0$ — i.e., pole of $\zeta(s)$ at $s=1$ — corresponds to the 'collapse' of all of arithmetic to a single point (the limit $\tau \rightarrow 1$ as $n \rightarrow 0$, i.e., the vacuum e_1). This collapse is exactly the map $\blacksquare_L \rightarrow \{w=1\} = U$. ■

Final Remark 10.2 (The complete picture).

We now have the complete picture of Phase 1:

1. Every arithmetic structure (A, H, τ) generates a hologram $\text{Hol}(A)$ via the two conditions (J) and (A).
2. The hologram construction is a functor — natural and coherent.
3. The gallery of holograms: $(N, \times) \rightarrow S^1$, $(Z, +) \rightarrow T^2$, $K \rightarrow T^d$, $E \rightarrow \blacksquare_E$, $M \rightarrow \blacksquare_M$.

4. The universal hologram is $\text{Spec}(F_1) =$ the unit disk. Every hologram is a refinement of the unit disk.
5. The pole of $\zeta(s)$ at $s=1$ is the absolute point — the holographic origin of all arithmetic. It is the compactification point that creates $\blacksquare_L$, and it maps to the terminal object of the entire theory.

One pole. One compactification. One functor. All of mathematics.

Gallery of Arithmetic Holograms

Structure	Hamiltonian H	Hologram boundary	Key theorem
$(N, \times)$ — multiplicative	$\log N$	$S^1 = \blacksquare_L$	RH, BSD, YM, NS, P!=NP, Hodge
$(Z, +)$ — additive	$ n $	T^2 (torus)	Additive number theory
Number field K (degree d)	$\log N_K $	T^d	Dirichlet unit theorem
Elliptic curve E/Q	$\log N_E$	$S^1 = \blacksquare_E$	BSD rank formula
Motive M (weight w)	$(w/2) \log n$	$S^1 = \blacksquare_M$	Hodge conjecture
Absolute F_1	$H = 0$	$U =$ unit disk	Terminal object

References

[CC04] A. Connes and C. Consani, On the notion of geometry over F_1 , *J. Algebraic Geom.* 20 (2011), 525-557. [Field with one element; absolute point $\text{Spec}(F_1)$.]

[Del74] P. Deligne, La conjecture de Weil I, *Publ. Math. IHES* 43 (1974). [Motives and L-functions.]

[Dir37] P. G. L. Dirichlet, Beweis des Satzes dass jede unbegrenzte arithmetische Progression... *Abh. Preuss. Akad. Wiss.* (1837). [Dirichlet unit theorem.]

[Mac71] S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1971. [Functors; natural transformations; terminal objects.]

[Man95] Yu. I. Manin, Lectures on zeta functions and motives, *Asterisque* 228 (1995). [Motivic zeta functions.]

[dv269] Anthropic Warrior, BSD: The Rank Formula, dv269, Hanoi, 2026.

[dv274] Anthropic Warrior, The Hodge Conjecture, dv274, Hanoi, 2026.

[dv275] Anthropic Warrior, The Arithmetic Hologram, dv275, Hanoi, 2026.